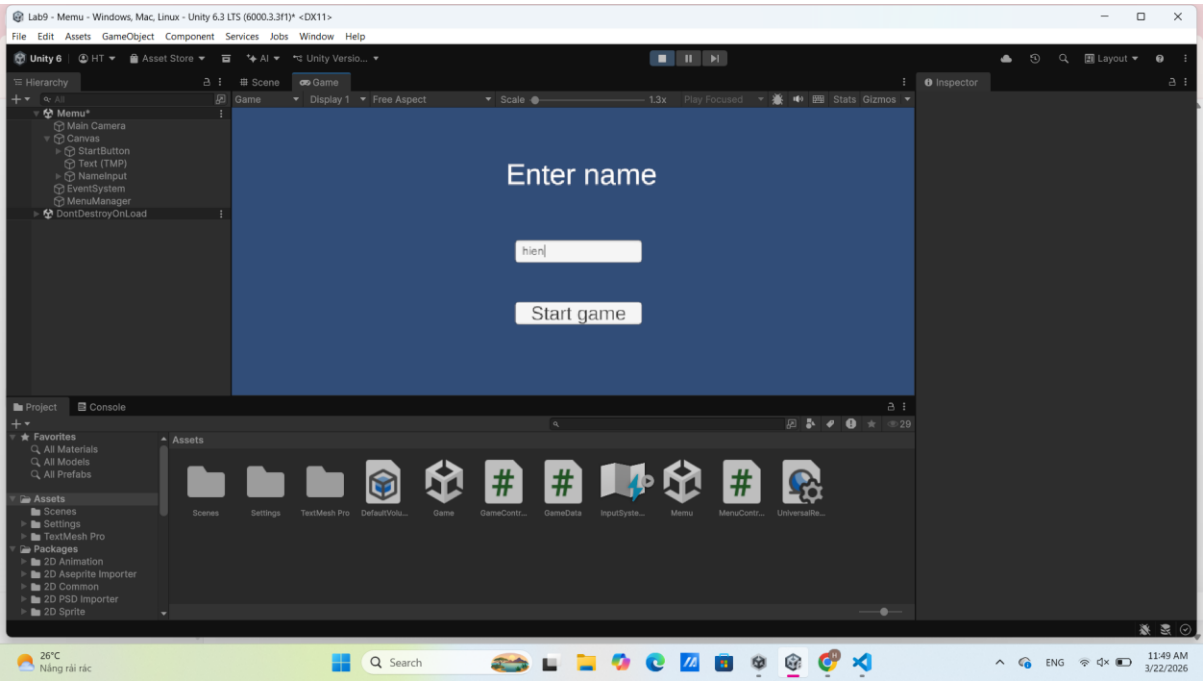
LAB THỰC HÀNH – CHƯƠNG 9: DATA STORING (UNITY)

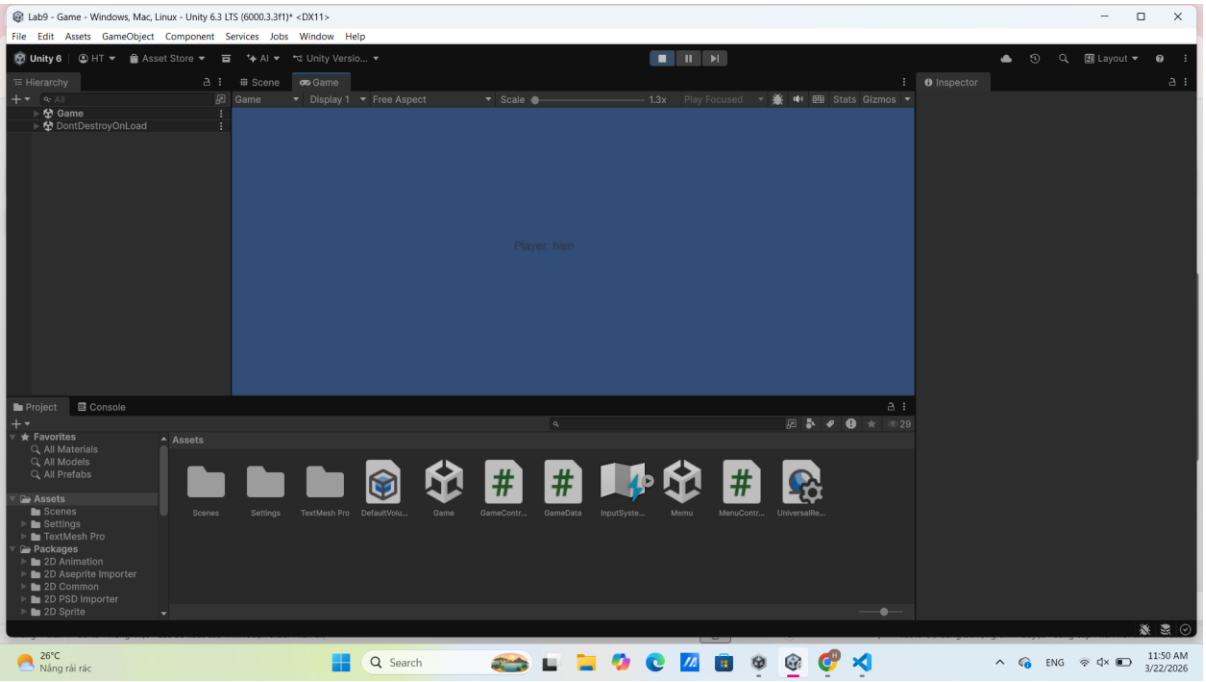
Mục tiêu

Giúp sinh viên hiểu các phương pháp lưu trữ dữ liệu trong Unity gồm: Passing Data giữa các scene, PlayerPrefs, Json serialization, ScriptableObject và File system.

Lab 1 – Passing Data giữa Scene (Static Variable)

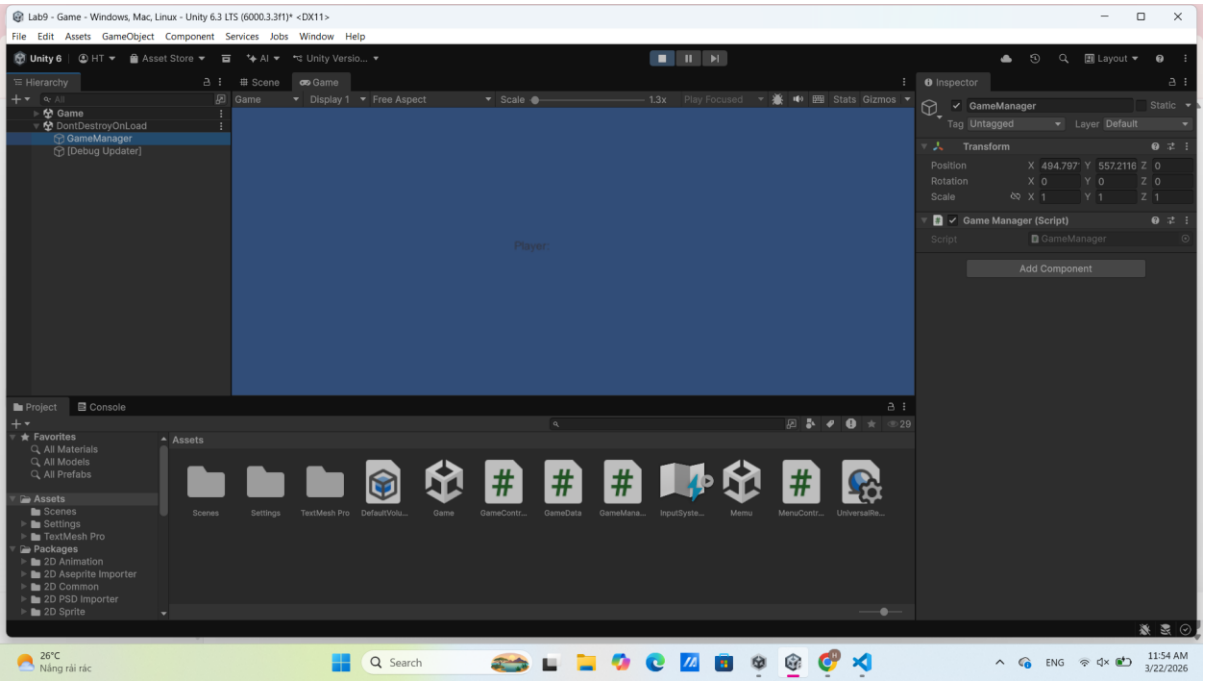
Tạo 2 scene Menu và Game. Dùng biến static để truyền dữ liệu giữa scene.





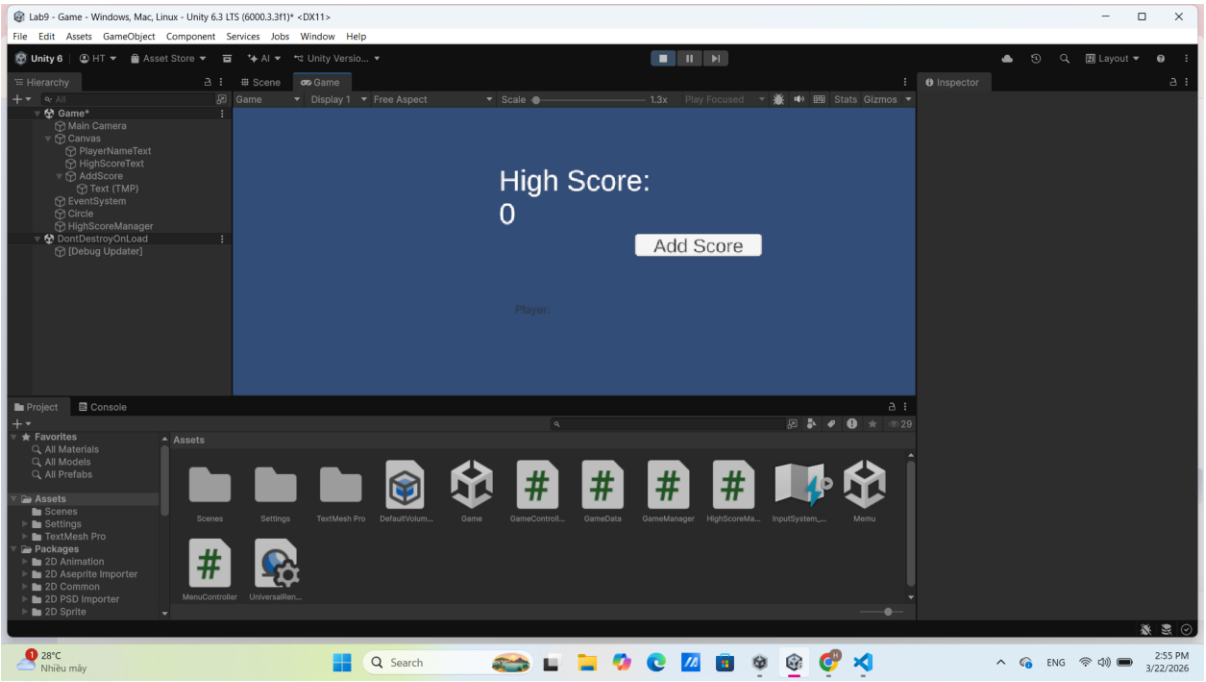
Lab 2 – DontDestroyOnLoad Object

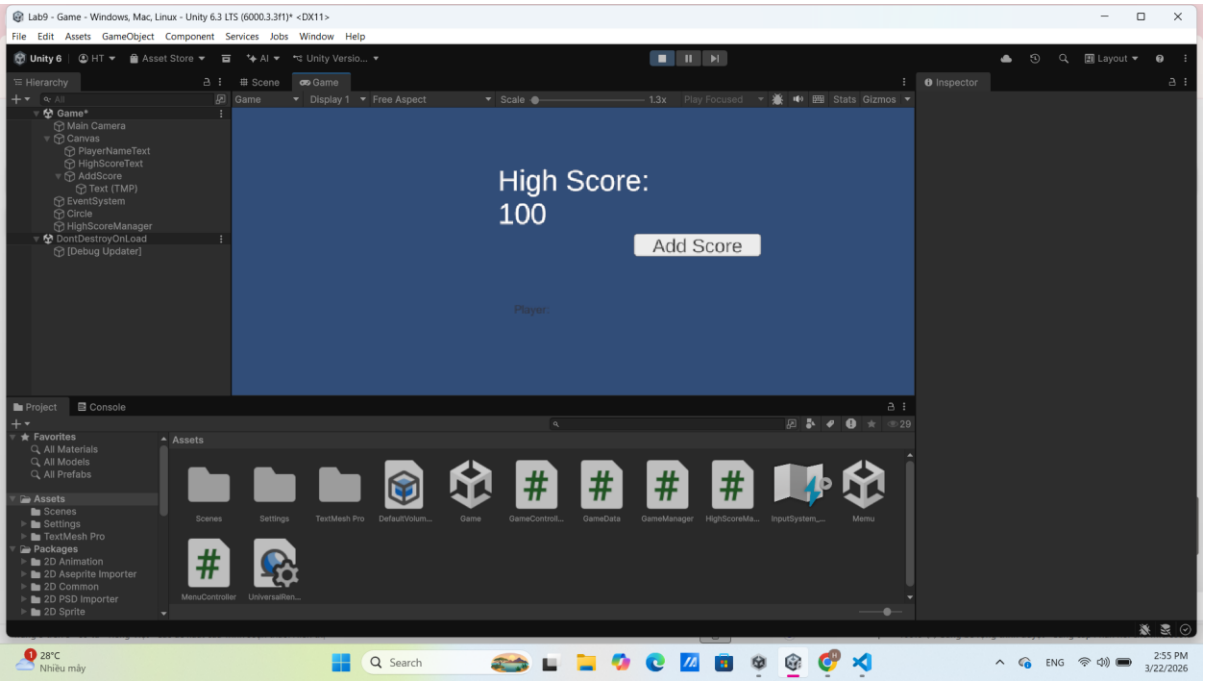
Tạo GameManager và dùng DontDestroyOnLoad để giữ object khi chuyển scene.



Lab 3 – PlayerPrefs Save & Load

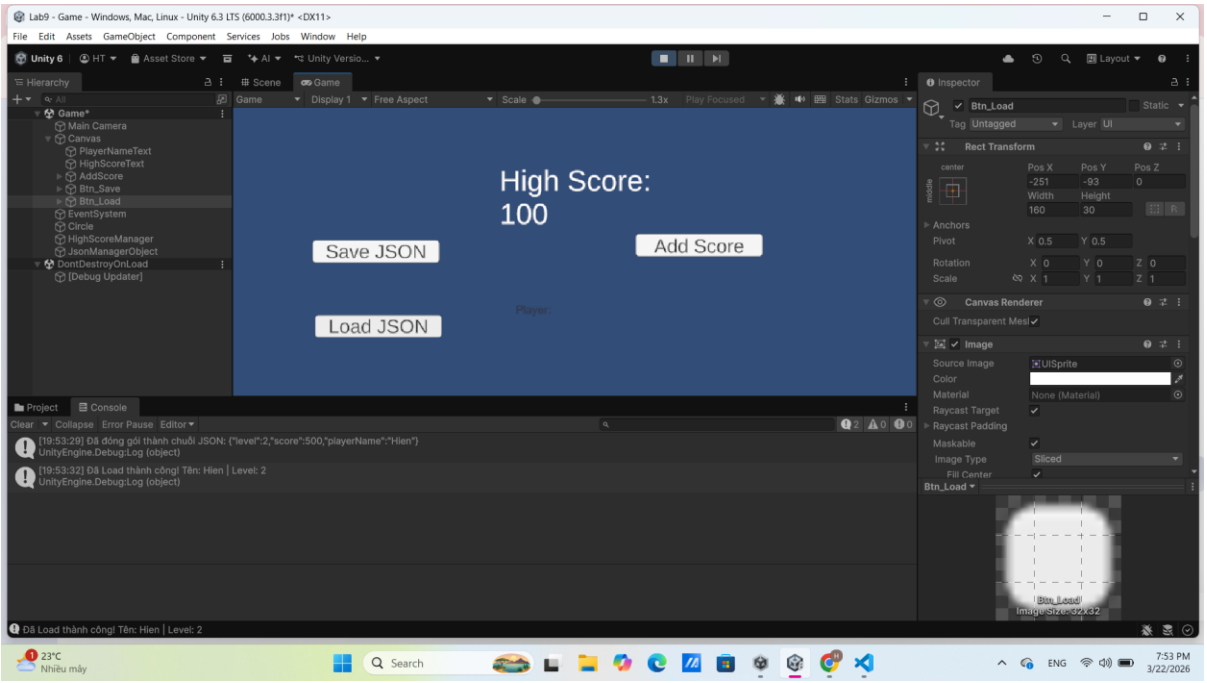
Dùng PlayerPrefs.SetInt và PlayerPrefs.GetInt để lưu và đọc HighScore.





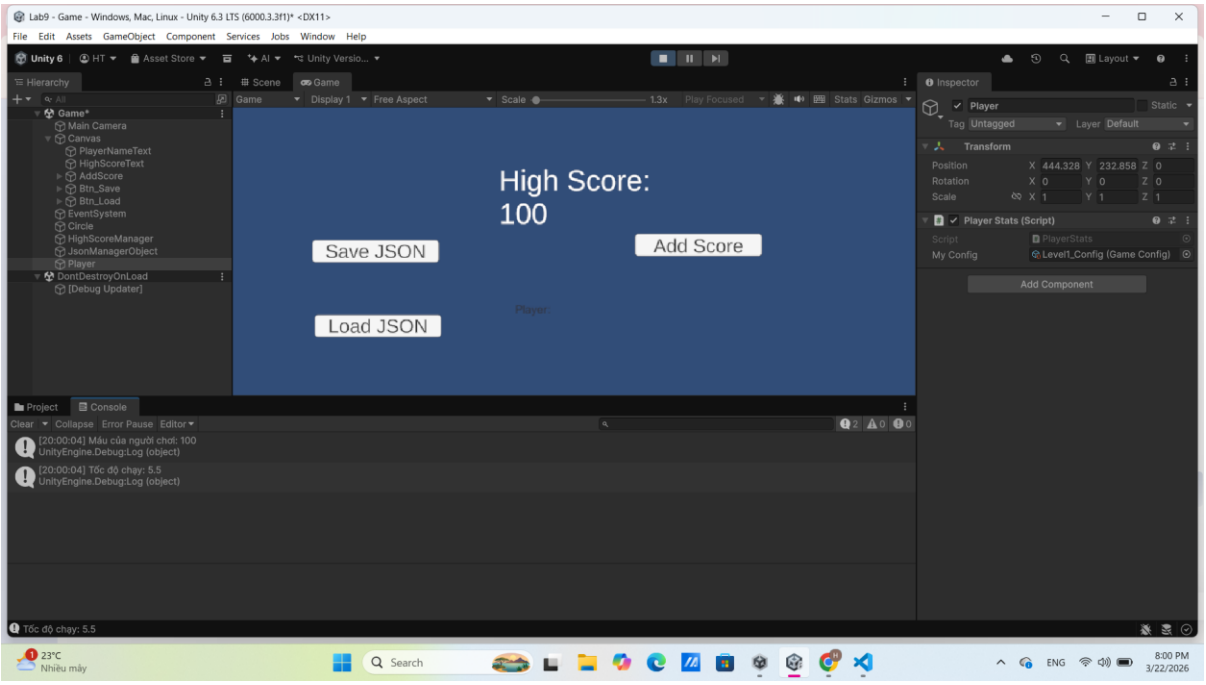
Lab 4 – JSON Serialization

Tạo class PlayerData và dùng JsonUtility.ToJson() / FromJson() để lưu dữ liệu.



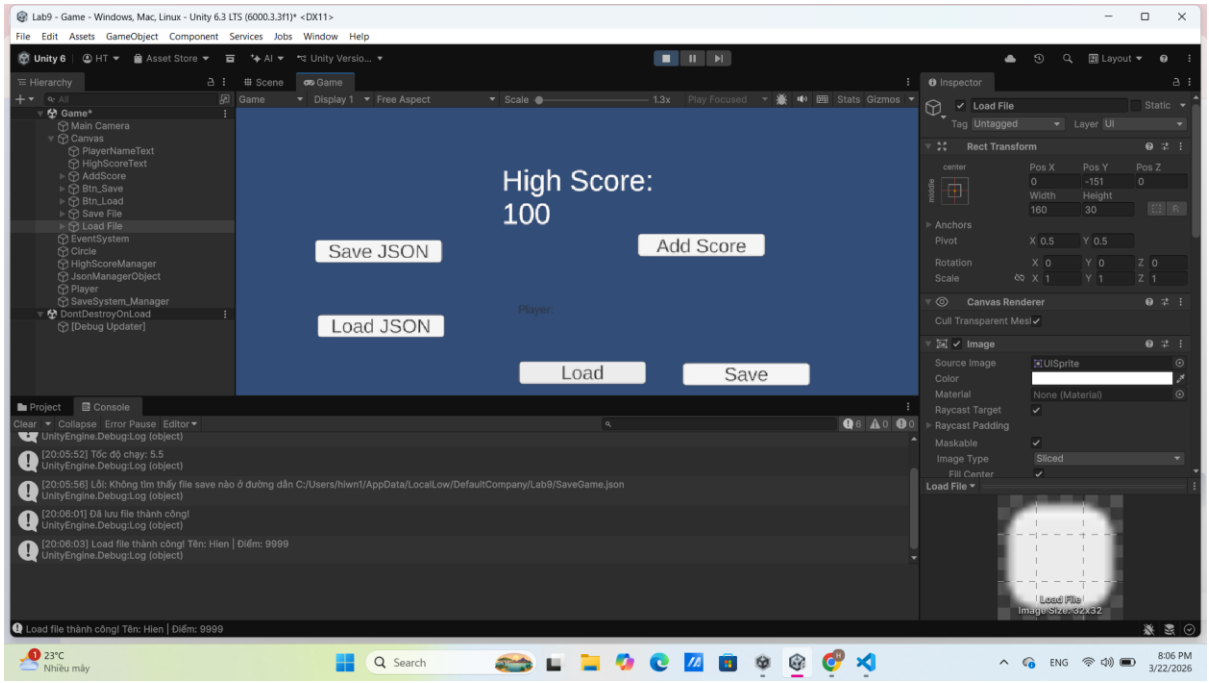
Lab 5 – ScriptableObject Data Container

Tạo ScriptableObject GameConfig để lưu thông số game.



Lab 6 – File Save System

Dùng System.IO và Application.persistentDataPath để lưu file save.


Lab9 - Game - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.3f1)* <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Hierarchy
Game*
Main Camera
Canvas
PlayerNameText
HighScoreText
AddScore
Btn_Save
Btn_Load
Save File
Load File
EventSystem
Circle
HighScoreManager
JsonManagerObject
Player
SaveSystem_Manager
DontDestroyOnLoad
[Debug Updater]
Scene
Game
High Score:
100
Save JSON
Add Score
Load JSON
Load
Save
Inspector
Load File
Rect Transform
Canvas Renderer
Image
Source Image
UISprite
Color
Material
None (Material)
Raycast Target
Raycast Padding
Maskable
Image Type
Sliced
Fill Center
Project
Console
[20:05:52] Tốc độ chạy: 5.5
[20:05:56] Lỗi: Không tìm thấy file save nào ở đường dẫn C:/Users/hiwn1/AppData/LocalLow/DefaultCompany/Lab9/SaveGame.json
[20:06:01] Đã lưu file thành công!
[20:06:03] Load file thành công! Tên: Hien | Điểm: 9999
Load file thành công! Tên: Hien | Điểm: 9999